

⇒ Important Formulas: MTH404

$$\textcircled{1} \quad \vec{v} = \frac{d\vec{r}}{dt}, \quad \textcircled{2} \quad \vec{a} = \frac{d^2\vec{r}}{dt^2}$$

and $\vec{a} = \frac{dv}{dt}$

⇒ Cartesian Components of Velocity and acceleration.

$$\textcircled{1} \quad \vec{v} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j}$$

⇒ Magnitude of Velocity

$$\textcircled{2} \quad |\vec{v}| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \frac{ds}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = \frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j}$$

⇒ Tangential and Normal Components of velocity & Acceleration

Velocity:-

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$$v = |\vec{v}| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \frac{ds}{dt}$$

where s is distance of particle along the path from some fixed point.

⇒ Normal component of velocity is always zero.

Acceleration :-

Tangential component of $\vec{a}$:-

$$\vec{a}_T = \frac{dv}{dt} \hat{t}$$

Normal component of $\vec{a}$:-

$$\vec{a}_N = \frac{v^2}{\rho} \hat{n}$$

where

ρ is radius of curvature

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left|\frac{dy}{dx}\right|}$$

$$\left|\frac{dy}{dx}\right|$$

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⇒ Radial and Transverse components of velocity and acceleration :-

Radial component of velocity :-

$$V_r = \frac{dr}{dt} = \dot{r}$$

Transverse component of velocity :-

$$V_\theta = r \frac{d\theta}{dt} = r\dot{\theta}$$

Radial/Transverse component of acceleration :-

$$\vec{a} = \left[\frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2 \right] \hat{r} + \left[2 \frac{dr}{dt} \frac{d\theta}{dt} + r \frac{d^2 \theta}{dt^2} \right] \hat{\theta}$$

which can be written as :-

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2) \hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta}) \hat{\theta}$$

and also

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2) \hat{r} + \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta}) \hat{\theta}$$

⇒ Radial Component of acceleration-

$$a_r = \ddot{r} - r\dot{\theta}^2$$

⇒ Transverse Component of acceleration

$$a_\theta = (2\dot{r}\dot{\theta} + r\ddot{\theta}) = \frac{1}{r} \frac{d}{dt} (r^2 \dot{\theta})$$

⇒ Distance Travelled by Particle in time t is -

$$x = ut + \frac{1}{2} at^2$$

⇒ Distance Travelled by particle in terms of velocity

$$2ax = v^2 - u^2$$

$$\left[\frac{dr}{dt} + r \frac{d^2\theta}{dt^2} \right] \hat{s}$$

⇒ Distance in n^{th} unit of time is

$$\text{Distance} = u + \frac{1}{2} a (2n-1)$$

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② Differential Equations of motion:-

⇒ Time dependent acceleration:-

$$\frac{d^2x}{dt^2} = a(t) \quad \text{or} \quad \frac{dv}{dt} = a(t)$$

⇒ Velocity dependent acceleration

$$v \frac{dv}{dx} = a(v) \quad \text{or} \quad \frac{dv}{dt} = a(v)$$

⇒ Distance dependent acceleration

$$v \frac{dv}{dx} = a(x) \quad \text{or} \quad \frac{d^2x}{dt^2} = a(x)$$

⇒ Velocity of Particle executing S.H.M:-

$$v = \sqrt{\lambda(a^2 - x^2)}$$

⇒ Distance covered by particle executing S.H.M

$$x = a \cos \sqrt{\lambda} t$$

⇒ Time Period of Particle executing S.H.M:-

$$T = \frac{2\pi}{\sqrt{\lambda}}$$

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$\Rightarrow$ Frequency of Particle executing S.H.M. is-

$$\nu = \frac{\sqrt{\lambda}}{2\pi}$$

Equation of motion of particle-

$$\frac{d}{dt}(mv) = F$$

$$m \frac{d\vec{v}}{dt} = \vec{F}$$

$$m\vec{a} = \vec{F}$$

Since $\dot{\gamma} = v$ and $\ddot{\gamma} = \vec{a}$

So

$$\vec{F} = m\ddot{\gamma}$$

$\Rightarrow$ Two Dimensional Polar Form of equation of motion-

• Radial component of force

$$\vec{F}_r = m(\ddot{\gamma} - \gamma(\dot{\theta})^2)$$

• Transverse component of force-

$$\vec{F}_\theta = m(2\dot{\gamma}\dot{\theta} + \gamma\ddot{\theta})$$

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For a particle moving along a circle of radius a with center O ,
 $\dot{r} = 0$ and $\ddot{r} = 0$

So

$$-F_r = ma(\ddot{\theta})^2, \quad F_\theta = ma\ddot{\theta}$$

Work:-

$$dW = \vec{F} \cdot d\vec{r}$$

Power:-

$$P = \frac{dW}{dt} = \vec{F} \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

Kinetic energy:-

$$T = \frac{1}{2}mv^2$$

Conservative field force:-

$$\vec{F} = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k}\right)$$

$$\vec{F} = -\nabla V$$

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where

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

→ Potential of force or Potential energy :-

• F is conservative if and only if $\text{curl } F = 0$

Total Energy :-

$$E = T + V$$

T = Kinetic energy.

V = Potential energy.

$$W = T_2 - T_1$$

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Best Regards
to Hāira Zafar